

PHYS 661 Fall 2024

Problem set 6, due November 26 2024. Each problem is 10 pts unless otherwise stated. Submit your homework either (1) via Brightspace or by (2) emailing it directly to the TA at gbowling@purdue.edu.

1 1D δ -potential revisited (15 pts)

Consider the 3D attractive potential $V(r) = \begin{cases} -V_0 & r < a \\ 0 & r > a \end{cases}$.

1. Solve the s-wave ($l = 0$) Schrodinger equation exactly and find the scattering phase shift $\delta_0(k)$ as a function of momentum $k = \sqrt{\frac{2m}{\hbar^2}E}$, V_0 , and a . (It will be a complicated expression, no need to simplify it.) Find an equation for k such that the phase shift vanishes $\delta_0(k) = 0$ (no need to solve the equation).
2. Find the equation for bound state energies E in terms of V_0, a (no need to solve the equation).
3. Give your best effort to verify Levinson's theorem, that the number of bound states is given by $N_b = \frac{1}{\pi}[\delta(E \rightarrow 0) - \delta(E \rightarrow \infty)]$.

2 Three angular momenta

Consider three angular momentum operators $\mathbf{S}, \mathbf{L}, \mathbf{I}$ in the $S = 1, L = 2, I = \frac{1}{2}$, representations, respectively. Consider also the operator $Q = a\mathbf{S} \cdot \mathbf{L} + b(\mathbf{S} + \mathbf{L}) \cdot \mathbf{I}$ where a, b are scalars.

1. What is the dimension of the matrix Q ? Show that the terms proportional to a, b in Q commute with each other.
2. Write Q in a block-diagonal form, in terms of a maximum set of commuting observables.

3 Some properties of rotation generators (15 pts)

1. Let L_x, L_y, L_z denote the components of the angular momentum operator. Show that if an operator O satisfies $[O, L_x] = [O, L_y] = 0$, then $[O, L_z] = 0$.

2. Let $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ denote the Pauli matrices, I the 2×2 identity matrix, and $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$. Show that

$$(\mathbf{a} \cdot \boldsymbol{\sigma})(\mathbf{b} \cdot \boldsymbol{\sigma}) = \mathbf{a} \cdot \mathbf{b} I + i(\mathbf{a} \times \mathbf{b}) \cdot \boldsymbol{\sigma} \quad (1)$$

3. We know that any complex 2×2 matrix M_2 can be written in terms of $I, \sigma_x, \sigma_y, \sigma_z$ as a linear combination $M_2 = a_0 I + \mathbf{a} \cdot \boldsymbol{\sigma}$. Show that any complex 3×3 matrix M_3 can be written in terms of I_3, S_x, S_y, S_z where I_3 denotes the 3×3 identity matrix and S_x, S_y, S_z the spin-1 matrices, but that in this case M_3 will not be a *linear* function of S_x, S_y, S_z . (*Hint*: Investigate the anticommutators $\{S_i, S_j\}$.) The spin 1 matrices are [in units $\hbar = 1$]

$$S_x = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, S_y = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}, S_z = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}. \quad (2)$$